

Define — Instantiate — Activate

A Semantic Value Chain for AI-Integrated Knowledge Management

Using the OMG BMM Server as a Working Example

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The Problem

Organizations invest heavily in tools and data — but struggle to turn that data into decisions.

Common failure modes:

- Tools are connected but use **different vocabularies** for the same concepts
- Data exists but has **no governance** — no versioning, no audit trail, no review process
- Beautiful knowledge graphs are built but **never activated** — no one queries them, no one acts on them

The classic ontology project failure: a beautifully governed but unused knowledge graph.

The Three-Layer Framework

A semantic value chain requires three distinct layers:

Layer	Purpose	Answers
1. Define	Vocabulary governance	What kinds of things exist? How do they relate?
2. Instantiate	Artifact creation & governance	What are the actual resources in this project?
3. Activate	Outcomes & value delivery	What decisions can we make from this data?

This maps onto the classic **schema / instance / use** distinction from information architecture, applied to an AI assisted OSLC linked data ecosystem.

Layer 1 — Define

The meaning layer: It establishes shared understanding before any data is created.

Two complementary mechanisms:

- **Ontology governance** (e.g., TopBraid EDG) — stakeholder review workflows, change history, version control of the vocabulary, multi-user authoring
- **OSLC ResourceShapes** — formalize the vocabulary as a REST API contract: required properties, cardinality, allowed values, UI metadata for creation dialogs

Without Layer 1: Layer 2 produces a connected but semantically incoherent graph — links exist but mean different things in different tools.

Layer 1 Example: BMM Vocabulary

The **bmm-server** defines the OMG Business Motivation Model 1.3 as an RDF ontology:

Ends (what to achieve)

- Vision
- Goal
- Objective

Means (how to achieve it)

- Mission
- Strategy, Tactic
- Business Policy, Business Rule

Influencers & Assessment

- Influencer (External/Internal)
- Assessment (SWOT)
- Potential Impact (Risk/Reward)

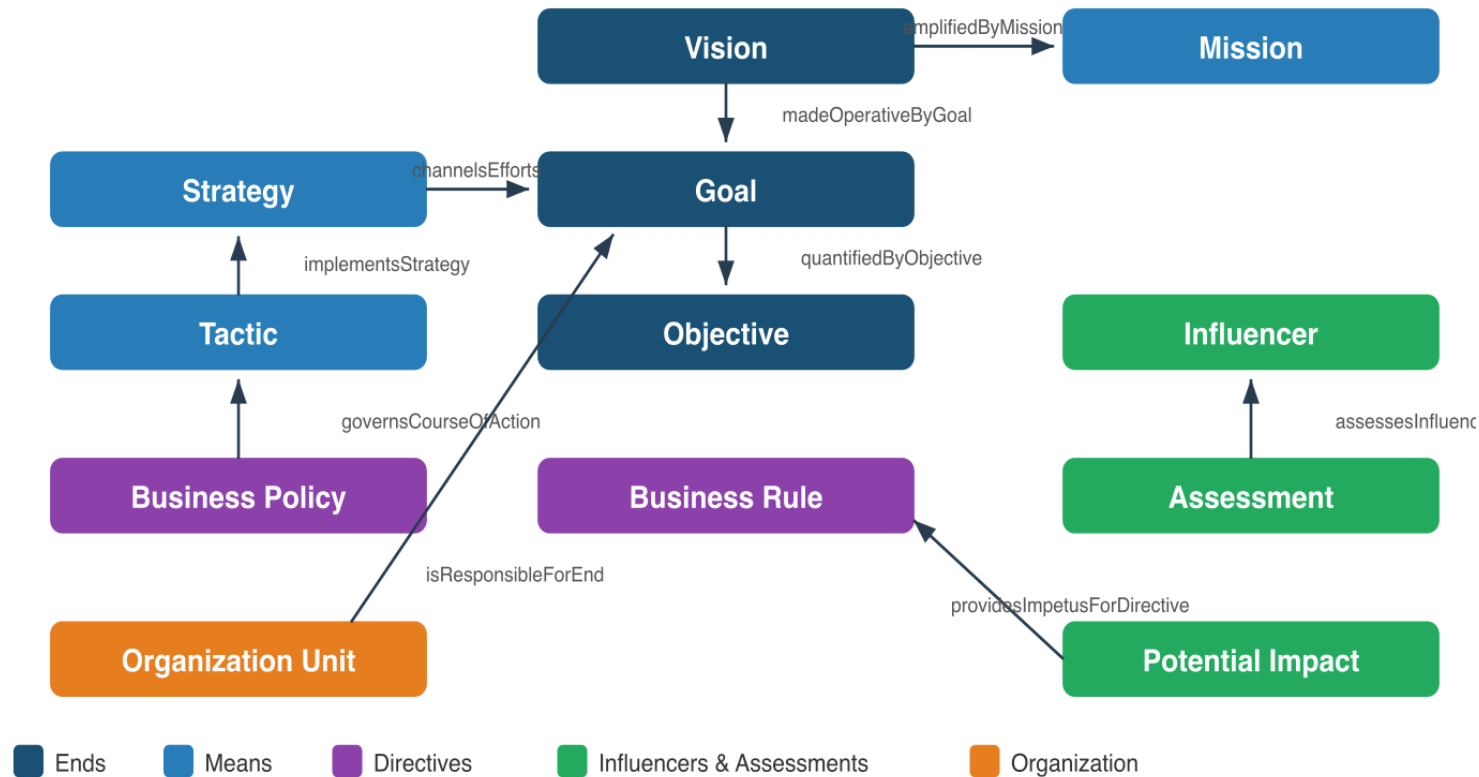
Organization

- Organization Unit
- Business Process
- Asset

25 classes, 49 properties, 14 ResourceShapes

Layer 1 Example: BMM Relationships

The ontology defines precisely how concepts connect. These typed relationships are what make queries, traceability, and AI analysis precise.



Layer 2 — Instantiate

The artifact layer: Here we transition from ontology experts to **subject matter experts** in the domain.

What it produces:

- Actual resources — requirements, plans, assessments, strategies
- Typed links between resources
- Version history and governance state (draft, approved, baselined)

Configuration management (streams, baselines, change sets) gives this layer its temporal dimension — reasoning about "the system as of this baseline" rather than just today's snapshot.

Without Layer 2 governance: Layer 3 can't answer versioned questions.

Layer 2 Example: EU-Rent

BMM Concept	EU-Rent Examples in Spec
Vision	1 — premium brand car rental
Goals	4 — premium brand, customer service, well-maintained cars, availability
Objectives	4 — A C Nielsen ratings, customer satisfaction, breakdown rate
Mission	1 — car rental across Europe and North America
Strategies	3+ — nationwide operation, car purchase/disposal, rewards scheme
Tactics	5+ — encourage extensions, outsource maintenance, standard specs, etc.
Business Policies	5+ — minimize depreciation, guarantee payments, no exports, etc.
Business Rules	6+ — match spec, lowest mileage, driver's license, service scheduling, etc.
Influencers	28+ — competitors, customers, regulations, technology, etc.
Assessments	6+ — SWOT: strengths, weaknesses, opportunities, threats
Potential Impacts	5+ — risks and rewards

All examples from the actual OMG specification, populated by AI reading the PDF.

AI Transforms Layer 2

Traditionally, Layer 2 was the bottleneck — entirely human-authored through forms and structured editors.

With MCP (Model Context Protocol), AI becomes a first-class participant:

1. **AI learns the domain model** from MCP resources (vocabulary, shapes, catalog)
2. **AI reads source documents** (specifications, plans, policy documents)
3. **AI identifies instances** — Visions, Goals, Strategies, Assessments
4. **AI creates and links resources** directly via the OSLC server API
5. **AI validates** against ResourceShape constraints

"Read the BMM 1.3 specification and create all the EU-Rent example artifacts and relationships described in the document."

Layer 3 — Activate

The value layer: Without it, Layers 1 and 2 produce a beautifully governed but unused knowledge graph.

Three activation mechanisms:

Mechanism	Use	Example
LQE - SPARQL/SQL	Analytical	Traceability reports, coverage metrics, validation
MCP Endpoint	Agentic	AI agents reasoning over live data, proposing changes
Tool Integrations	Operational	Engineers seeing linked data in DOORS Next, EWM, Polarion

Layer 3 Example: MCP Endpoint

The bmm-server exposes an MCP endpoint at `/mcp` with **34 dynamically generated tools**:

14 create tools

- one per BMM resource type

14 query tools

- one per BMM resource type

6 generic tools

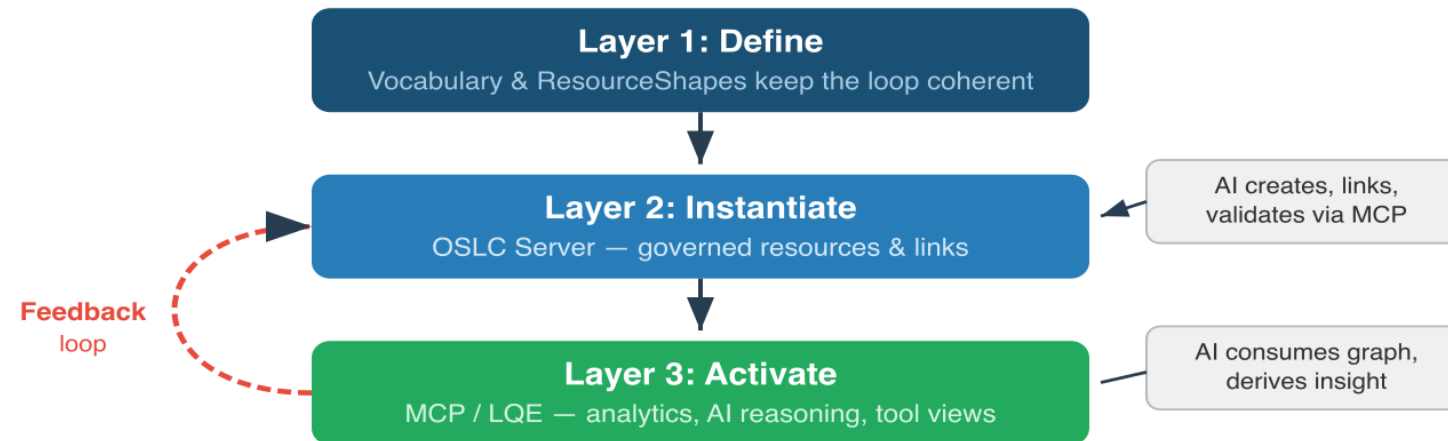
- `create_service_provider`
- `get_resource`, `update_resource`
- `delete_resource`
- `list_resource_types`, `query_resources`

3 MCP resources (for AI learning)

- `oslc://vocabulary`
- `oslc://shapes`
- `oslc://catalog`

The Feedback Loop

This creates a virtuous cycle that didn't exist before MCP:



Why Not Just Use AI Alone?

"Can't we just feed all our documents to an LLM and ask it questions?"

Yes — but AI outputs are ephemeral. A conversation produces text, not governed artifacts.

Concern	AI Alone	AI + OSLC Server
Audit trail	"Claude said so"	Versioned artifact with provenance
Persistence	Different answer next month	Living model with change history
Interoperability	Prose output	Machine-consumable linked data (RDF)
Governance	Chat session	Review workflows, access controls, sign-off
Precision	Fluent non-answers	Visible, queryable gaps
Repeatability	Statistical approximation	Deterministic queries on governed data

AI Needs Structure to Be Reliable

- **Better patterns, better results.** RDF assertions governed by ResourceShapes are consistent in expression, precisely typed, and richly linked.
- **The ontology gives the AI a map.** Without it, the AI is a very expensive search engine that produces fluent but structurally ungrounded answers.
- **Explicit gaps vs. hallucination.** The system of record forces explicit representation of what is known vs. unknown. A gap in the model is a visible, queryable gap — not a fluent non-answer.
- **Quantitative analytics.** Ontology-structured data delivers precise, repeatable results for compliance reporting and impact analysis.

What AI Brings to the System of Record

Authoring acceleration

SMEs who can't write RDF or navigate complex tool UIs can now contribute their knowledge conversationally. The AI translates intent into ontology-conformant resources.

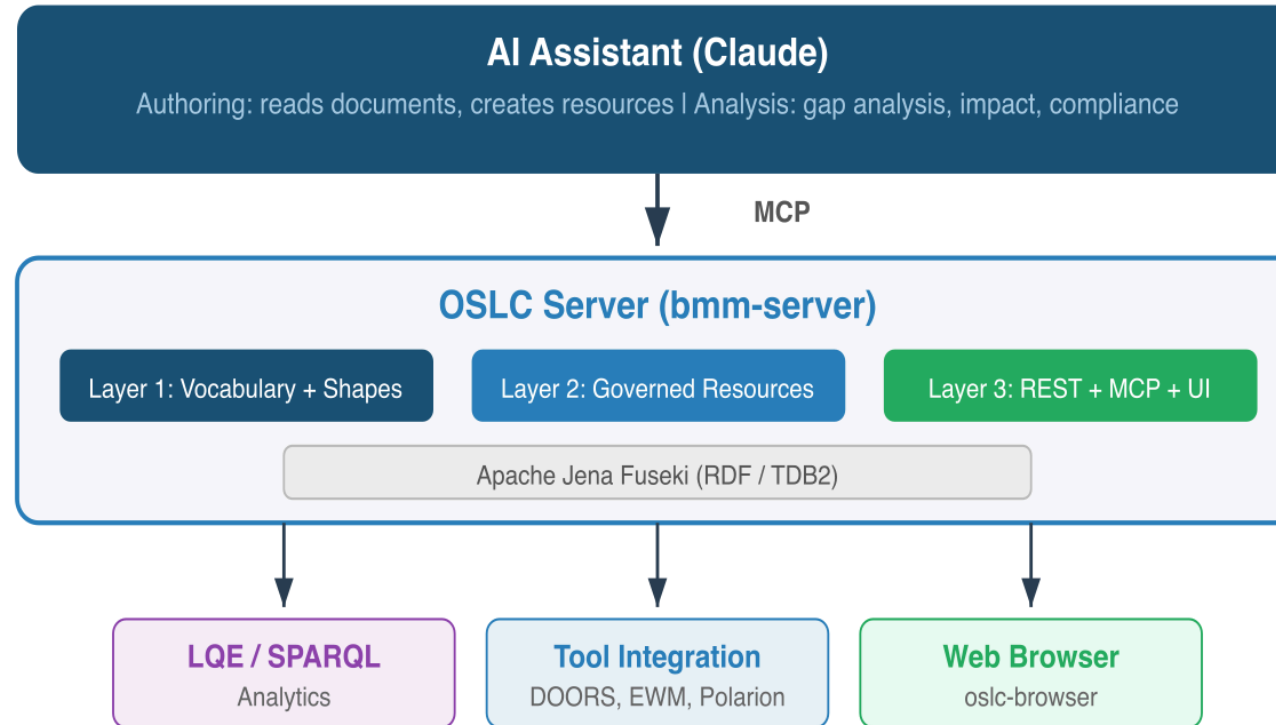
Analytical depth

AI can consume the entire linked data graph and perform analysis impractical for humans with queries and reports alone — identifying gaps, contradictions, and inconsistencies across hundreds of interconnected resources.

Humans in the loop

Ontologies provide stakeholder viewpoints — structured perspectives tailored to different roles. These keep humans meaningfully engaged, which matters because humans take responsibility for action and outcome.

The Integrated Architecture



BMM Server: A Complete Working Example

Aspect	Implementation
Domain	OMG Business Motivation Model 1.3
Layer 1	25 classes, 49 properties in BMM.ttl; 14 ResourceShapes
Layer 2	RDF triple store (Jena Fuseki); EU-Rent example from BMM 1.3 spec, populated by AI
Layer 3	OSLC REST API + MCP endpoint (34 tools) + oslc-browser UI
AI Integration	MCP endpoint exposes vocabulary/shapes for learning; tools for CRUD
Port	localhost:3005

Try it: Start Fuseki, then `cd bmm-server && npm start`

AI prompt: *"Read docs/BMM-formal-15-05-19.pdf and create all the EU-Rent example artifacts and relationships from the spec."*

Key Takeaway

The **Define-Instantiate-Activate** framing positions ontologies and OSLC servers not as alternatives to AI, but as the infrastructure that makes AI-assisted work:

- **Auditable** — every resource has provenance and version history
- **Repeatable** — deterministic queries on governed data, not statistical approximation
- **Governable** — review workflows, access controls, multi-stakeholder sign-off
- **Interoperable** — machine-consumable linked data across tools and organizations

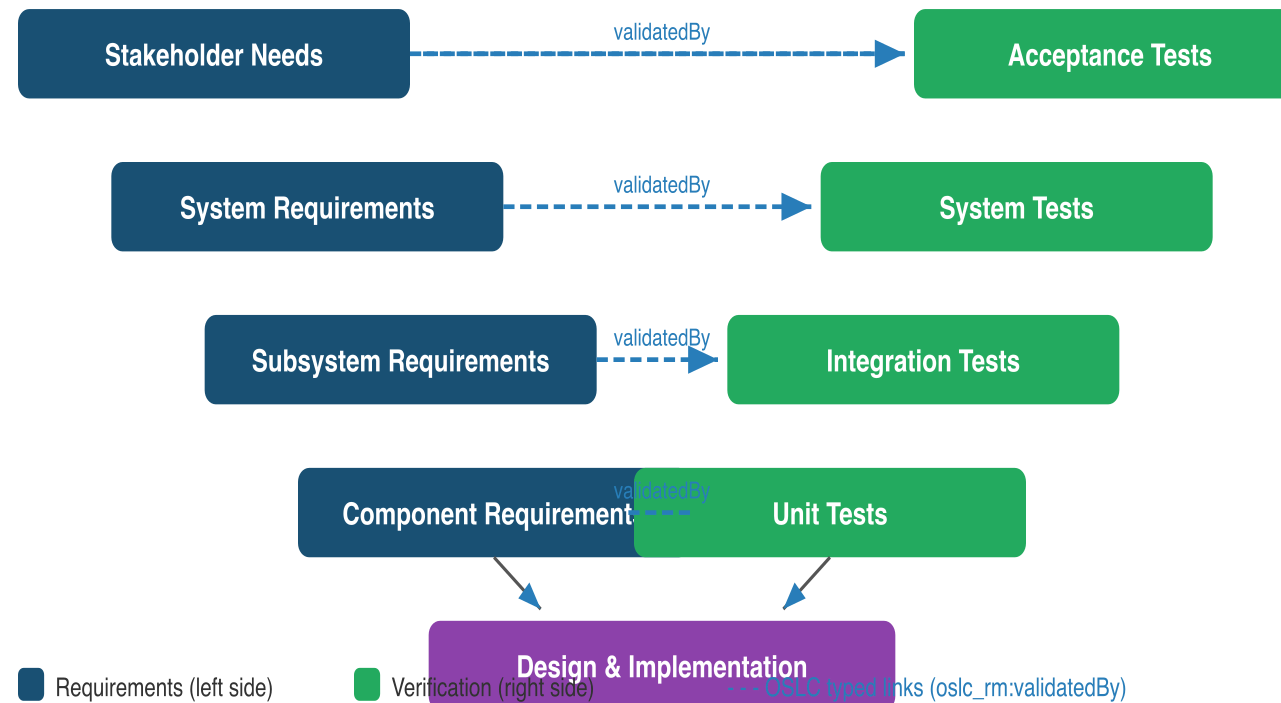
*The OSLC server is the **system of record**.*

*The AI is the most capable **authoring and analysis tool** that system of record has ever had.*

*The ontology is what makes their collaboration **semantically precise** rather than statistically approximate.*

Applying Define-Instantiate-Activate to an AI-Assisted V-Model

The framework applies not just to individual OSLC servers, but to the **entire systems engineering lifecycle**.



In OSLC terms, each traceability link is **typed** — the V-model's traceability is a **live link graph** spanning tools

Three Layers of AI Assistance

An AI assistant could be connected via MCP to an integrated tool chain operates at three layers:

Layer	Scope	MCP Access	Example
1. Tool-Local	Single tool	Tool's own MCP	DOORS Next AI improves requirement quality
2. Integration	Cross-tool	OSLC server MCP	"Which requirements lack test cases?"
3. Analytics	Lifecycle-wide	LQE/TRS MCP	Coverage ratios, compliance, impact analysis

Layer 1 improves authoring within each tool silo.

Layer 2 enables cross-tool reasoning over the OSLC link graph.

Layer 3 provides efficient read-only analytics across the entire lifecycle.

Scenario: Requirements Change Impact

An engineer changes a system requirement: performance threshold from 100ms to 50ms.

Phase 1 — Impact Discovery (Layer 3, LQE)

AI queries the materialized graph for full downstream impact.

Result: "3 subsystem requirements, 12 component requirements, 8 test cases (2 passing, 3 draft, 3 missing), 4 work items affected"

Phase 2 — Triage and Planning (Layer 2, OSLC)

AI traverses live links to assess each affected artifact. Flags test cases needing updates. Identifies pre-existing coverage gaps made urgent by the change.

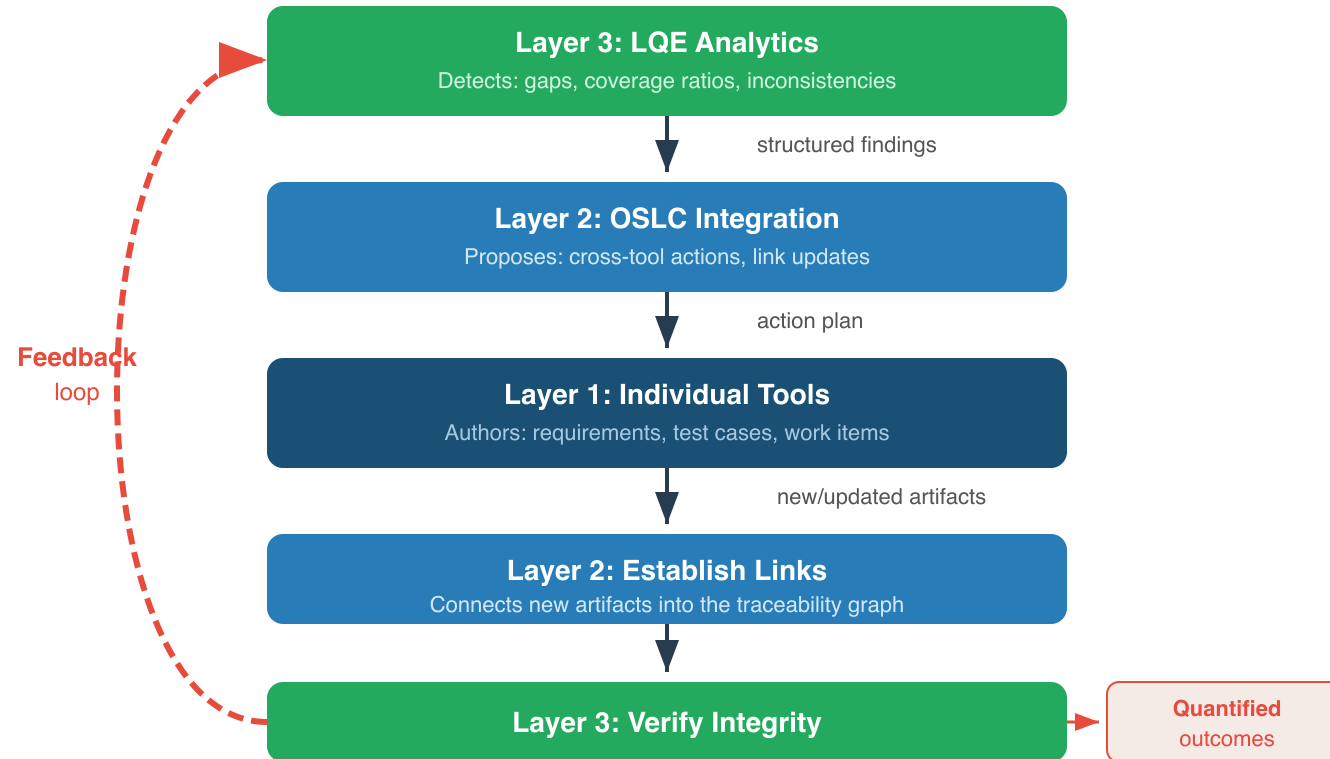
Phase 3 — Assisted Authoring (Layer 1, Tools)

ETM: drafts updated test procedures. DOORS Next: proposes revised subsystem allocations. EWM: creates linked change requests.

Phase 4 — Verification (Layer 3, LQE)

AI re-queries to confirm all gaps closed, coverage restored.

The V-Model Feedback Loop



This is **Define-Instantiate-Activate** applied to the lifecycle: vocabularies define valid traceability, tools instantiate artifacts and links, analytics activate the data — feeding back into new versions.

Governance: Authority and Approval

Governance helps ensure we achieve intended outcomes with proper authority and approval traceability.

Level	AI Action	Approval	Example
Observe	Query and report	None needed	LQE gap analysis, coverage reports
Propose	Draft artifacts in "Draft" state	Human review required	AI-generated test cases, requirement updates
Execute	Create links by policy	Pre-authorized	Mechanical linking: test case to requirement

The AI operates within OSLC access controls — it does not bypass governance. The AI authenticates with the user's identity; the same role-based permissions apply whether the request comes from a browser or from an AI through MCP.

Governance: Traceability and Metrics

Traceability of AI actions — Every AI action records provenance: what triggered it, what analysis justified it, what policy authorized it, what human approved it. TRS propagates these records to LQE.

Quantifiable outcomes:

Metric	What It Measures
Coverage ratio	Requirement-to-test traceability before and after
Gap closure rate	Gaps resolved per cycle
Change propagation completeness	Downstream artifacts updated within time window
Consistency scores	SHACL validation against V-model structural rules
Cycle time	Requirement change to verified traceability closure

*These metrics measure the **engineering process**, not the AI. The AI makes the process faster and more complete.*

Thank You

Resources:

- oslc4js repository: github.com/jamsden/oslc4js
- BMM Server: [oslc4js/bmm-server/](https://github.com/jamsden/oslc4js/tree/master/bmm-server)
- Define-Instantiate-Activate: [oslc4js/docs/Define-Instantiate-Activate.md](https://github.com/jamsden/oslc4js/blob/master/docs/Define-Instantiate-Activate.md)
- OSLC specifications: open-services.net
- OMG BMM 1.3: [omg.org/spec/BMM/1.3](https://www.omg.org/spec/BMM/1.3)